ACADEMIC MCQ

Kamani Science & Prtaprai Arts College
Amreli

2024 - 2025

Developed by :
Nirmeet Makwana

Project Report on

Online Examination System

Kamani Science & Prtaprai Arts College, Amreli

Developed By

Nirmeet Makwana (003303225506)

Guided By

Mr. Paras Thummar

Year Of Submission

Submitted in partial fulfillment of the requirement for the award of the degree of Bachelor of Computer Application (BCA)

2023 - 2024



Kamani Science College
&
Prataprai Arts College



Vidyavihar,Bhavnagar Road,Amreli-365601

C E R T I F I C A T E

This is to certify that

Mr./Miss. Makwana Nirmeet Navnitbhai

Roll. No 5072 Studying in B.C.A. Semester V has completed his/her Project work satisfactorily during the year 2024-2025.

Signature Project in Charge	Signature Head of Department	Signature Examiner

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Preface

The main objective of any computer science student is to get practical knowledge as possible. Being able to have a practical knowledge by developing a project is a lifetime experience. As practical knowledge is important as theoretical knowledge we are thankful of having a project.

Through the development of the project we had a great experience of various strategies that can be applied in development of the project. This project is the stepping stone for our career.

I am pleased to present this project report. Proper case has been taken while organizing the report so that it is easy to comprehend. Also, various software engineering concepts have been implemented.

Acknowledgment

I am student of BCA sem-5 Studying in the Kamani Science & prtaprai Arts College Amreli. We Thanks to the entire person who has given their Support in shaping of the system.

I am thank Mr. Paras Thummar for giving us guidance and co-operation in understanding the system. We also thanks them for their unconditional help in making of this project.

I am have great deal of gratitude towards our head of department who encourages in taking up this activity. We thank all faculties and administrative staffs of the institute.

I am also thankful to all those friends who have helped me in this endeavor either of indirectly & especially my family.

Thanks to all!!

Introduction

Now days, The advent of digital technology has revolutionized various sectors , Including education. One significant innovation is the development of online examination systems, Which offer a modern approach to conducting and managing assessments

This project is provide platform for exams and introduces a comprehensive online examination system designed to streamline the examination process for educational institutions. The system is structured into two pannel

There are two pannel:

Admin Pannel :-

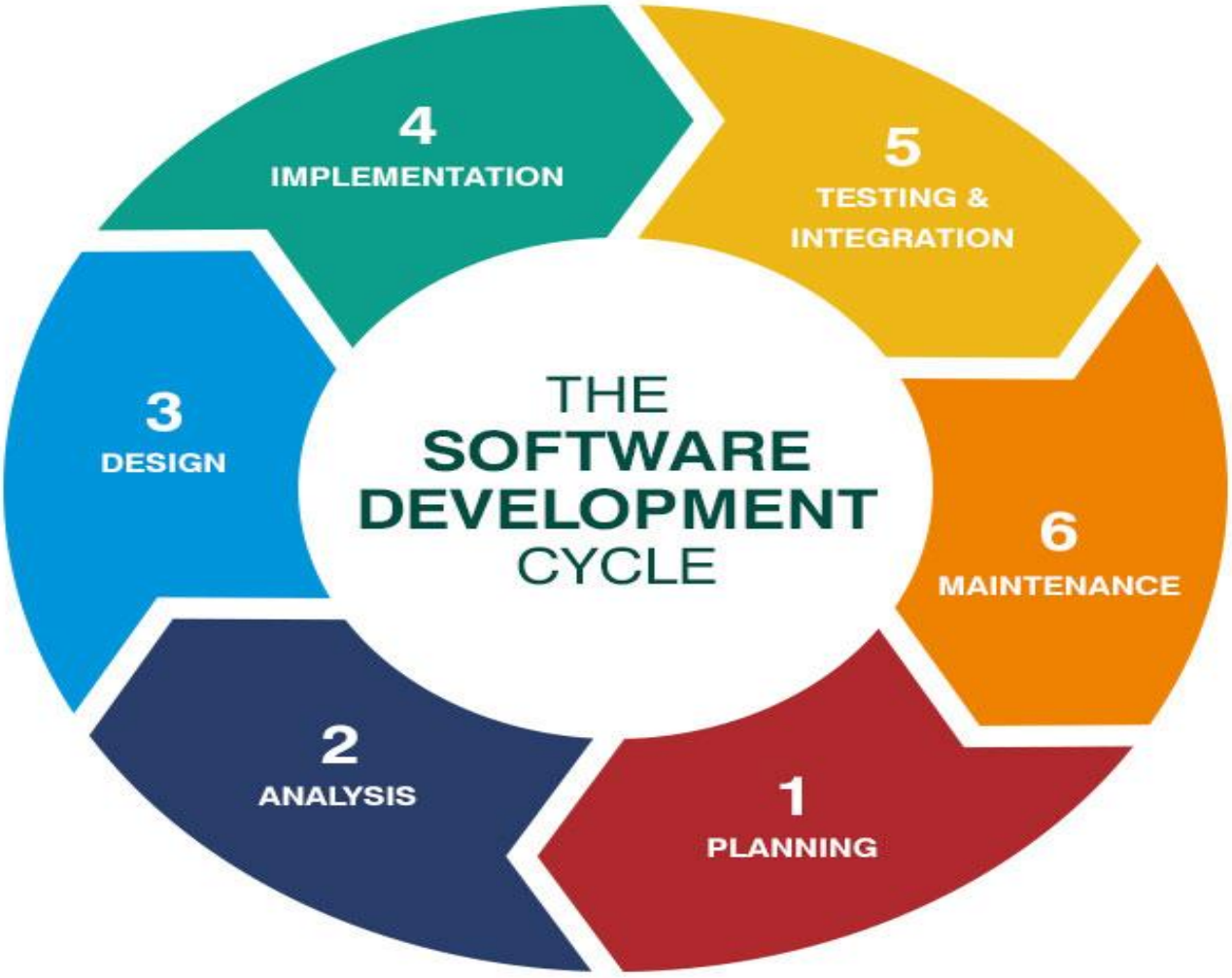
In this panel you can add class, subject and test to user.

User Pannel :-

In this panel user can access all the test posted by admin.

SDLC – Software Development Life Cycle

Software development life cycle is a structured process that is used to design , develop and test good – quality software. SDLC is a methodology that defines the entire procedures of software development step by step.



Requirements Specification

1. What is the purpose website

- The purpose of a website for an online Examination system is to facility the creation , management and administration of exams over the internet.

2. What type of exams will the system support?

- MCQ types of exams will the system support.

3. Who will be the primary users of the system?

- I . Techers (School Admin)
  II . Student(User)

4. Will the system support time-bond exams?

- Yes , with adjustable time for each exam

5. How will users authenticate them selves to access the system?

- Through unique usernames and password or single sing-on.

6. Who is add a new test ?

- A school admin or school teachers is add a new test.

7. How to add a new test?

- When someone from the school like teacher or admin login, thet will have menu bar in which theire will be an option of "Add Test" that help them to add the test.

8. How will the system ensure the security and integrity of the exams?

- By , using secure login , encrypted data , and monitoring for suspicious activity .

9. Will the system support time-bound exams?

- Yes , with adjustable time limits for each exam.

10. How will students receive their results?

- Immediately after the exam for objective types, or after manual grading for subjective types

11. What is role of school admin?

- The school admin role is add new test, see Students details, see students marks and which students attend test?

12. How many pannel in this website?

- There are tow pannels:
    i. School Admin pannel
    ii. Student pannel

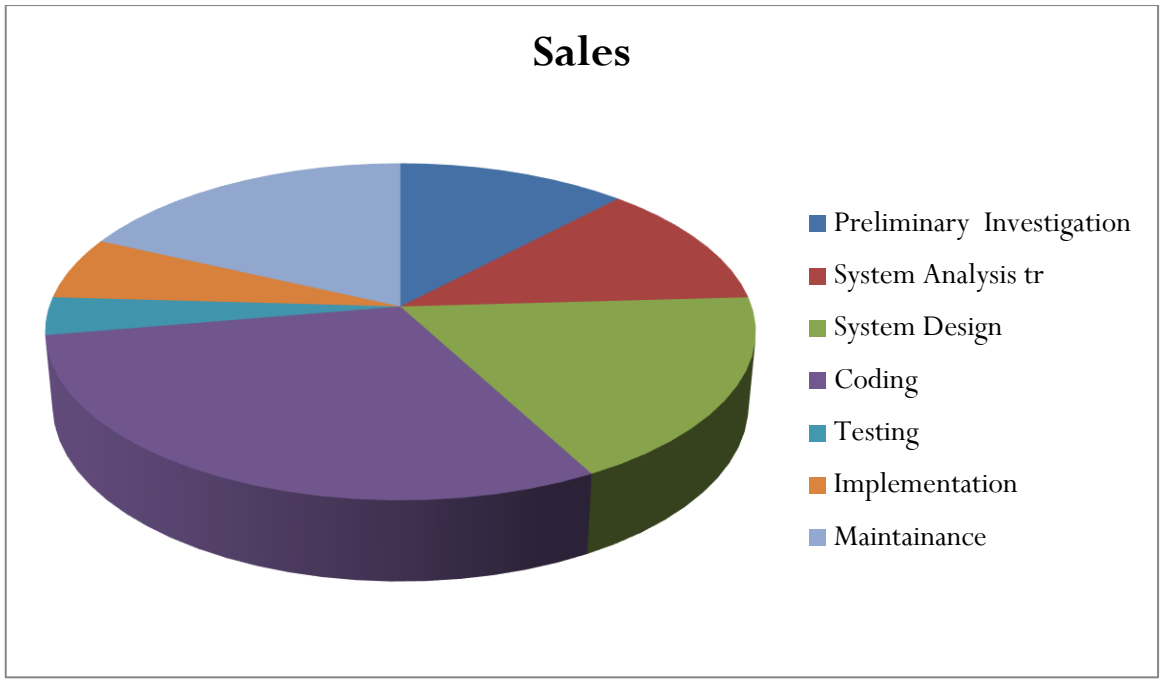
Project Problem Definition

- Add test Questions and Answer
- Student Registration
- Display test
- Check test

Project Profile

- **Project Name** : Online Examination System
- **Developed by** : Nirmeet Makwana
- **Front End tool** : Php,HTML,Java,script,JQuery,CSS
- **Back End tool** : PHP Mysql
- **Oprating System** : Microsoft Windows 7 & 10
- **Project Details** : Provide best platform for schools and students. The school Admin post/add test and students get this test, and attend this test.
- **Guided by** : Mr. Paras Thummar
- **Submitted by** : Saurashtra University, Rajkot

Time Line Chart



Preliminary Investigation	⇨	**10 Days**
System Analysis	⇨	**10 Days**
System Design	⇨	**15 Days**
Coding	⇨	**25 Days**
Testing	⇨	**5 Days**
Implementation	⇨	**5 Days**
Maintenance	⇨	**15 Days**

Hardware Requirement

- 512+ MB Of RAM
- Multimedia Keyboard And Mouse
- GHz Processor
- Printer
- Monitor

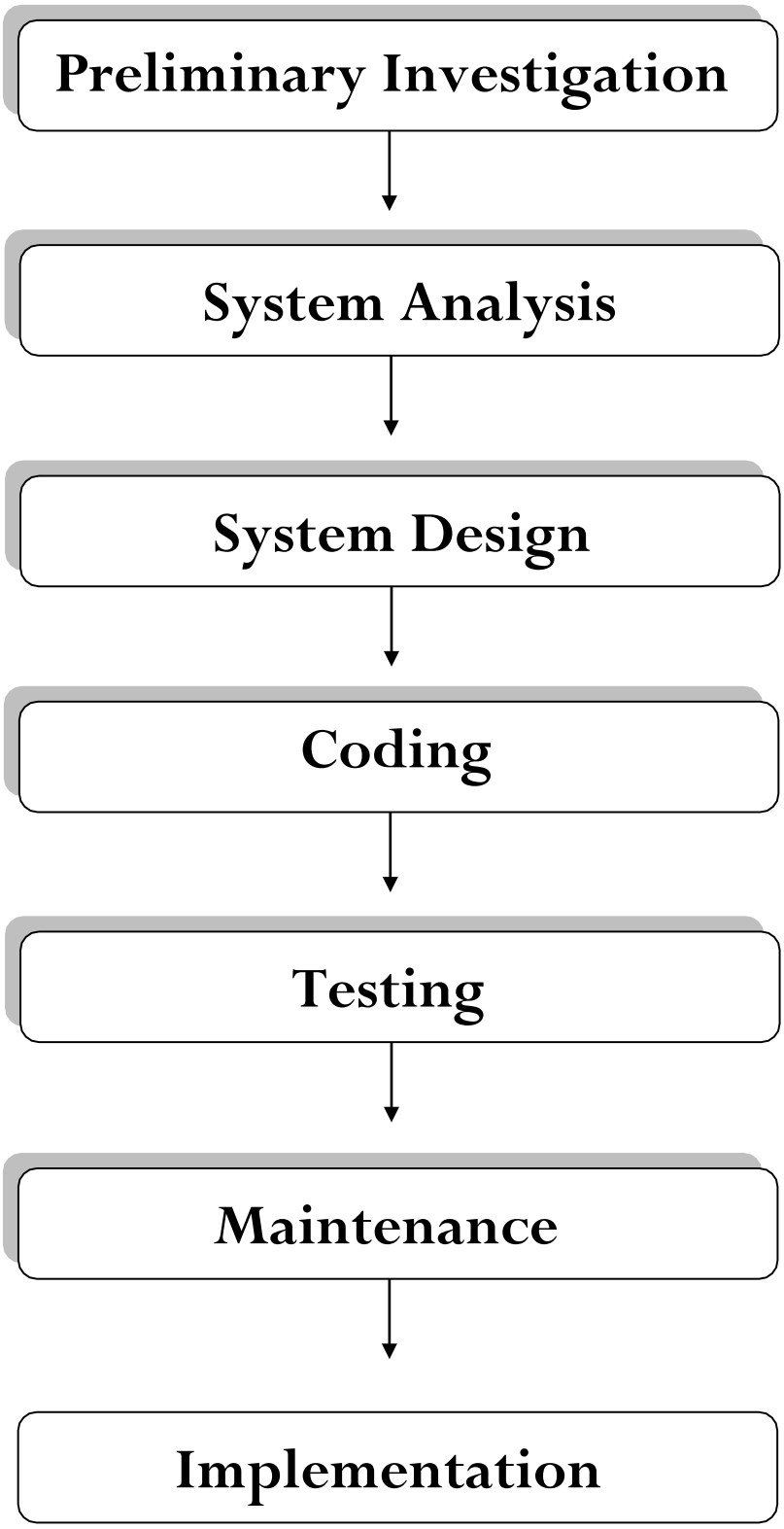
Software Requirement

- Operating System : Microsoft windows 7 & Microsoft windows 10
- Front End Tools : PHP ,HTML, CSS , Ajax , Visual Studio
- Back End Tools : PHP MySQL
- Documentation Tool : Microsoft word 2007
- Browser Specification : Google Chrome , Mozilla Firefox

Project Planning & Scheduling

Project planning is an aspect of Project Management, which comprises of various processes. The aim of these processes is to ensure that various Project tasks are well coordinated and they meet the various project objectives including timely completion of the project. Project Planning is an aspect of Project Management that focuses a lot on Project Integration. The project plan reflects the current status of all project activities and is used to monitor and control the project.

The Project Planning tasks ensure that various elements of the Project are coordinated and therefore guide the project execution. Project Scheduling is one of the most important task of Project Planning and also the most difficult tasks. In very large projects it is possible that several teams work on developing the project. They may work on it in parallel. However their work may be interdependent.


Preliminary Investigation
System Analysis
System Design
Coding
Testing
Maintenance
Implementation

System Analysis

System analysis is the process of identification of objectives and requirements, evaluation of alternative solutions and recommendation for a more feasible solution. In other words, system analysis is the step by step process of gathering, recording and interpreting facts. The main aim of analysis is to determine problem areas and to take solutions to reduce or eliminate them.

- **System analysis itself breaks into two stages:**
  1. **Preliminary analysis :**
  2. **Detailed analysis :**

During analysis, data are collected from the available files, decision points and transactions of the systems using various tools like Data Flow Diagram (DFD).

Feasibility Study

A purpose of feasibility study is to check out the possibility of a computerized solution to the organization's observed problem before very much money that has been spent on.

A feasibility study is carried out to select the best system that meets performance requirements. Only by spending the time to evaluate the feasibility do I reduce the chances for extreme embarrassment at later stage of the system project.

For the complete feasibility study I need to concentrate on following area:

- **They are three types of feasibility Study :**

1. Technical

2. Operational

3. Economical

❖ Technical

Technical feasibility considers whether the desired project can be completed within the framework of available technology. As our project was website designing, this was not much of a problem, as many advanced web-editing tools are available.

❖ Operational

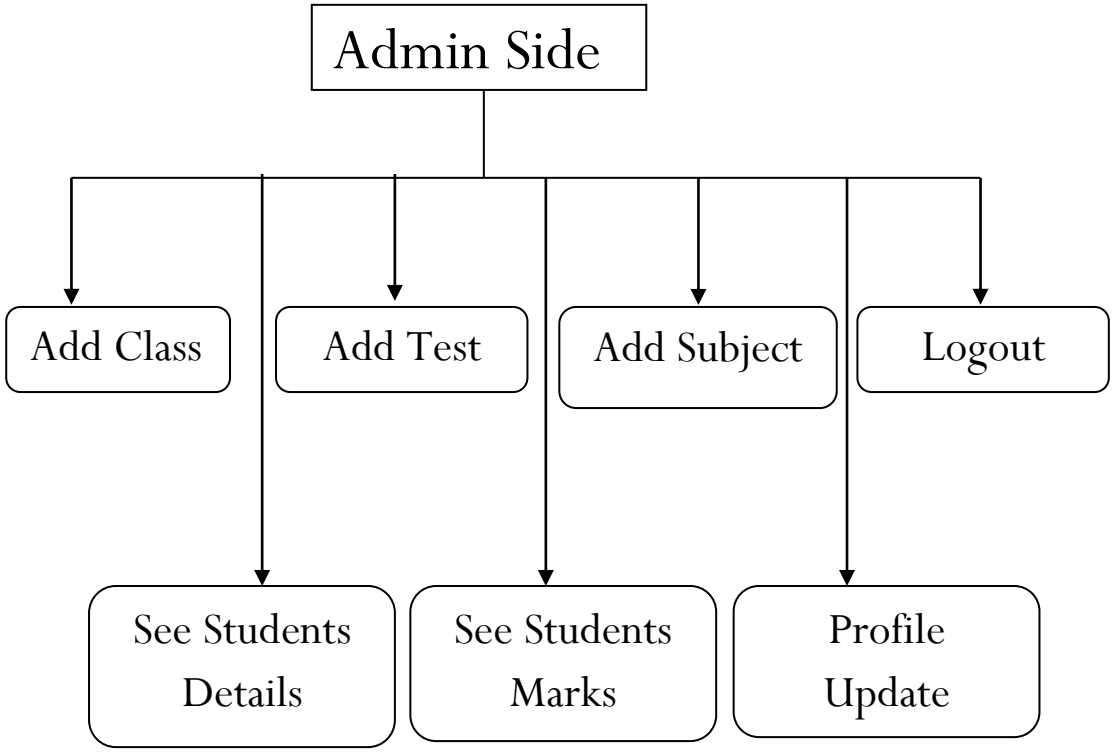
Operational feasibility was done to assure that the product that would be developed would be used. Since a cards and gifts have to have a website of its own sooner or later, that wasn't much of a problem.

❖ Economical

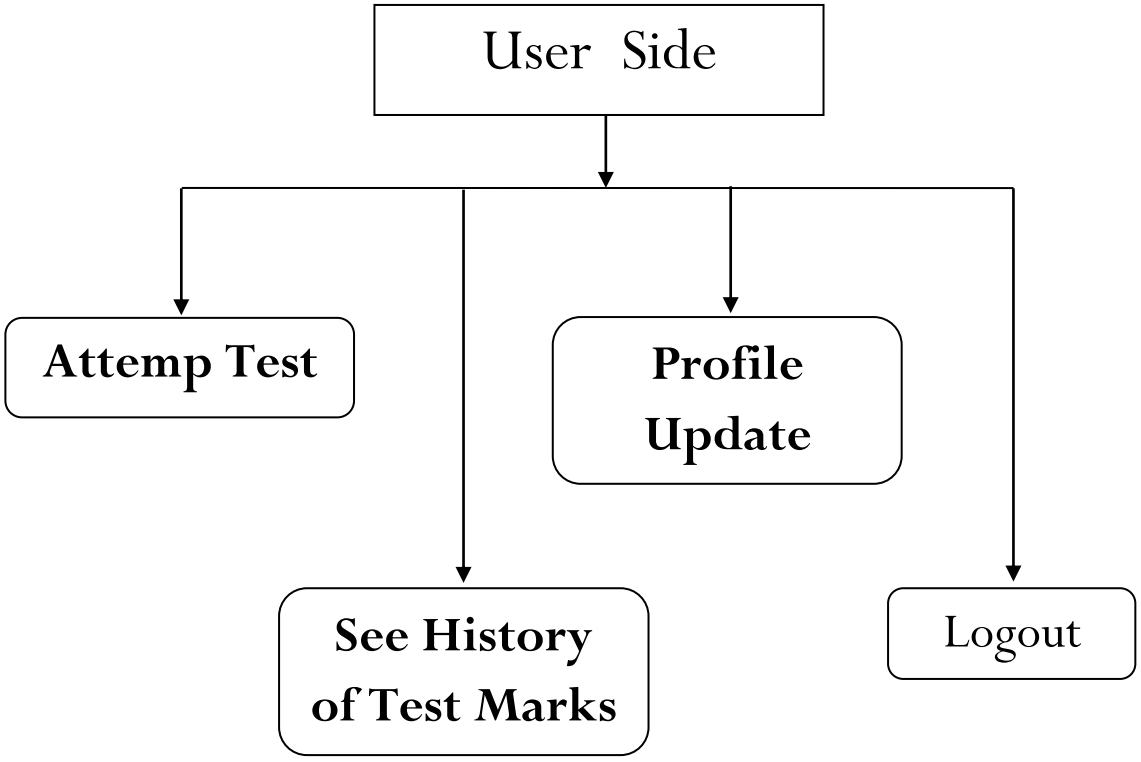
Our Commercial Website is Economically Feasible Because. We have developed this website only for one time. Only future enhancement is required on the other hand every client can view our site any time or many time as per their requirement. So the future economical expenses are null. So we can say that this website is totally economically feasible.

Project Page Structure

Admin Side :-



User Side :-


User Side
Attemp Test
Profile Update
See History of Test Marks
Logout

Data Dictionary

The Data Dictionary is a set of table . MySQL uses it to maintain information about this own database.

- **Data dictionary contains the following information:**
  - ✓ Name of the table of database.
  - ✓ User information such as privileges.
  - ✓ Name and data-type of all columns in database table.

- **Database Name : academic**

1. School :- This Table is used to store School information.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	S_ID	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	S_Name	varchar(170)	utf8mb4_general_ci		No	None			Change Drop More
3	S_username	varchar(40)	utf8mb4_general_ci		No	None			Change Drop More
4	S_Pass	varchar(12)	utf8mb4_general_ci		No	None			Change Drop More
5	S_Mobile	varchar(15)	utf8mb4_general_ci		No	None			Change Drop More
6	S_email	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
7	S_Date	varchar(25)	utf8mb4_general_ci		No	None			Change Drop More
8	S_address	varchar(150)	utf8mb4_general_ci		No	None			Change Drop More
9	S_logo	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More

2. Student :- This Table is used to store students information.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	U_Id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	u_fullname	varchar(70)	utf8mb4_general_ci		Yes	NULL			Change Drop More
3	u_name	varchar(40)	utf8mb4_general_ci		Yes	NULL			Change Drop More
4	U_Pass	varchar(12)	utf8mb4_general_ci		Yes	NULL			Change Drop More
5	U_Mobile	varchar(15)	utf8mb4_general_ci		Yes	NULL			Change Drop More
6	U_email	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More
7	S_ID	int(11)			Yes	NULL			Change Drop More
8	C_Id	int(11)			Yes	NULL			Change Drop More
9	u_address	varchar(150)	utf8mb4_general_ci		Yes	NULL			Change Drop More
10	U_date	varchar(25)	utf8mb4_general_ci		Yes	NULL			Change Drop More
11	U_photo	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change Drop More

3. Class :- This Table is used to store class.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	C_Id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	C_Name	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
3	S_Id	int(11)			No	None			Change Drop More

4. Subject :- This Table is used to store subject.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	sub_Id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	Subject_Name	varchar(35)	utf8mb4_general_ci		No	None			Change Drop More
3	S_Id	int(11)			No	None			Change Drop More

5.test_info :- This Table is used to store Test Information detail.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	T_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	T_Name	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
3	No_question	int(11)			No	None			Change Drop More
4	T_Limit	int(11)			No	None			Change Drop More
5	T_Marks	int(11)			No	None			Change Drop More
6	T_date	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
7	Sub_id	int(11)			No	None			Change Drop More
8	C_Id	int(11)			No	None			Change Drop More
9	S_Id	int(11)			No	None			Change Drop More

6. test_que_ans:- This Table is used to store test Question & Answer.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	Q_Id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	question	text	utf8mb4_general_ci		No	None			Change Drop More
3	A	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More
4	B	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More
5	C	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More
6	D	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More
7	Answer	varchar(50)	utf8mb4_general_ci		Yes	NULL			Change Drop More
8	T_Id	int(11)			Yes	NULL			Change Drop More
9	S_ID	int(11)			Yes	NULL			Change Drop More

7. t_marks:- This Table is used to store class

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	M_Id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	Marks	int(11)			Yes	NULL			Change Drop More
3	T_Id	int(11)			Yes	NULL			Change Drop More
4	u_id	int(11)			Yes	NULL			Change Drop More

.

Data Flow Diagram

The DFD gives brief idea on how the designed system is working. It also suggests to us the type of users who using this system and the process involved in the system.

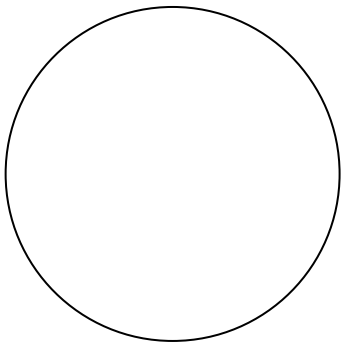
Represent data flow



Represent an Entity



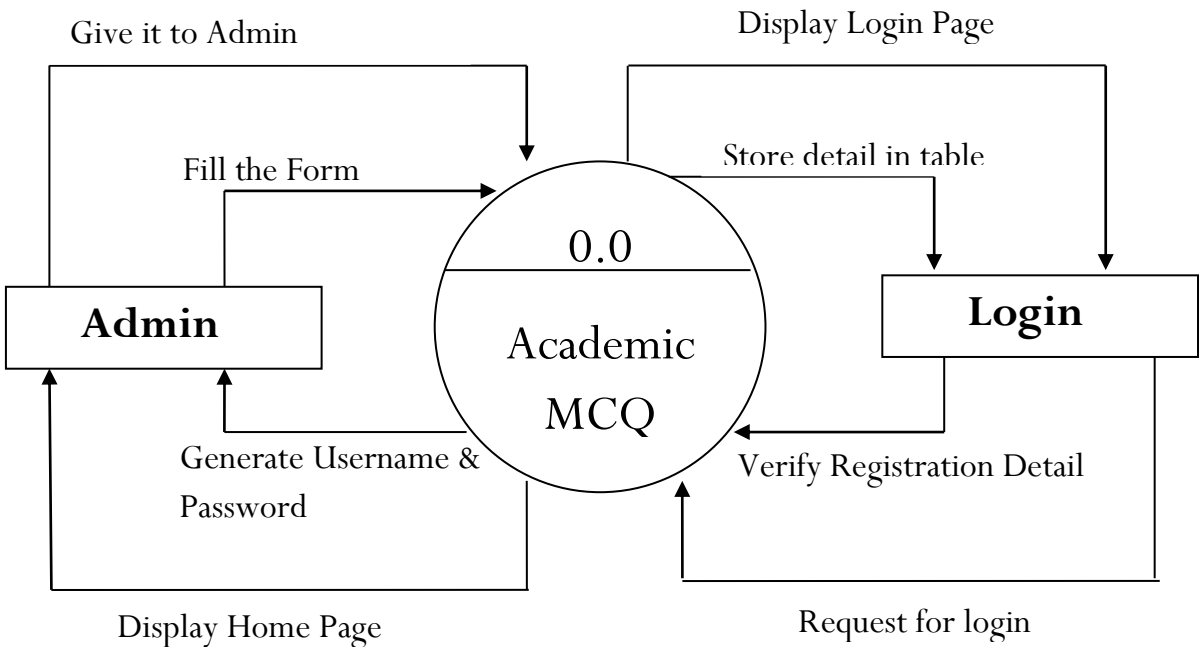
Represent a data table



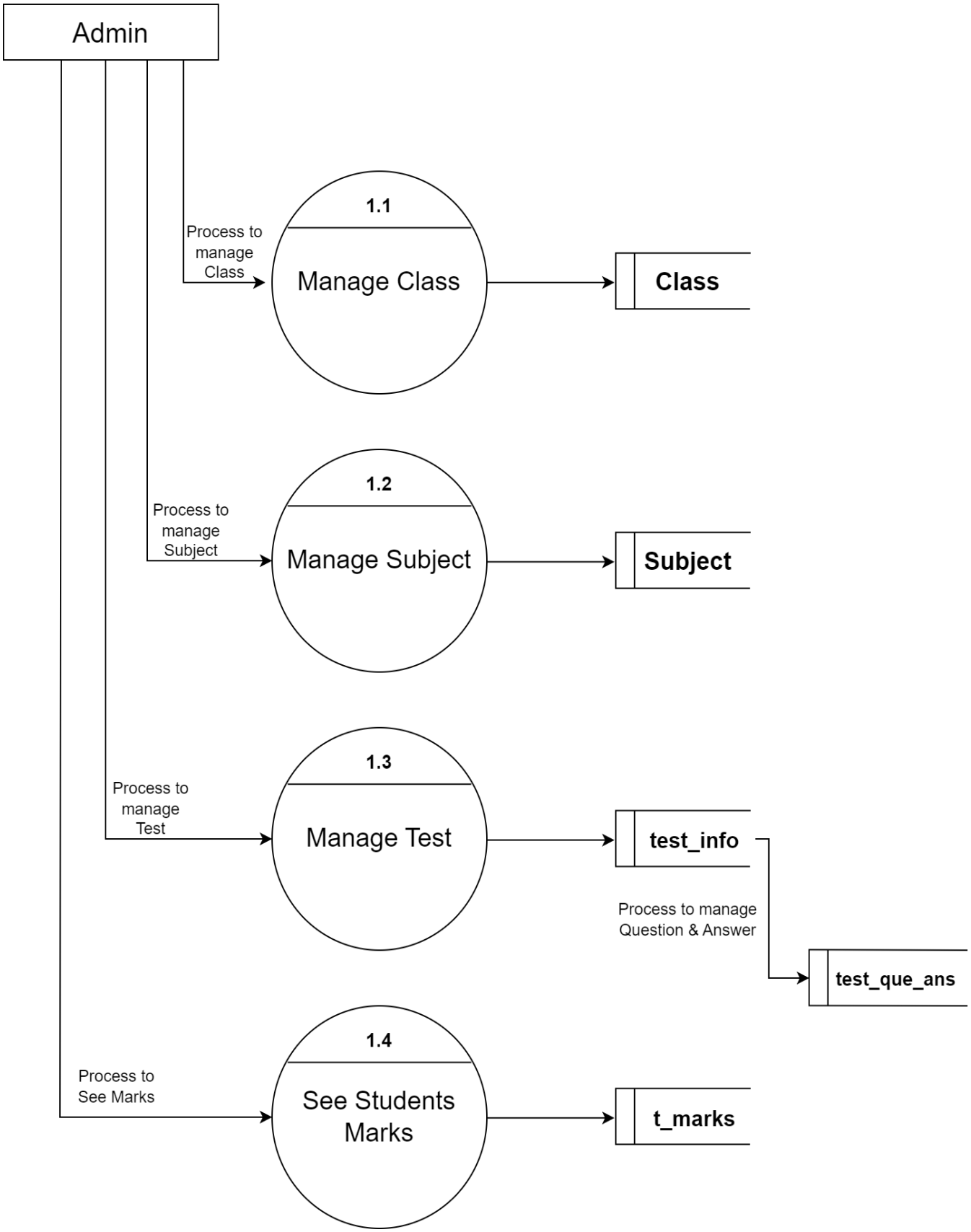
Represent a processing

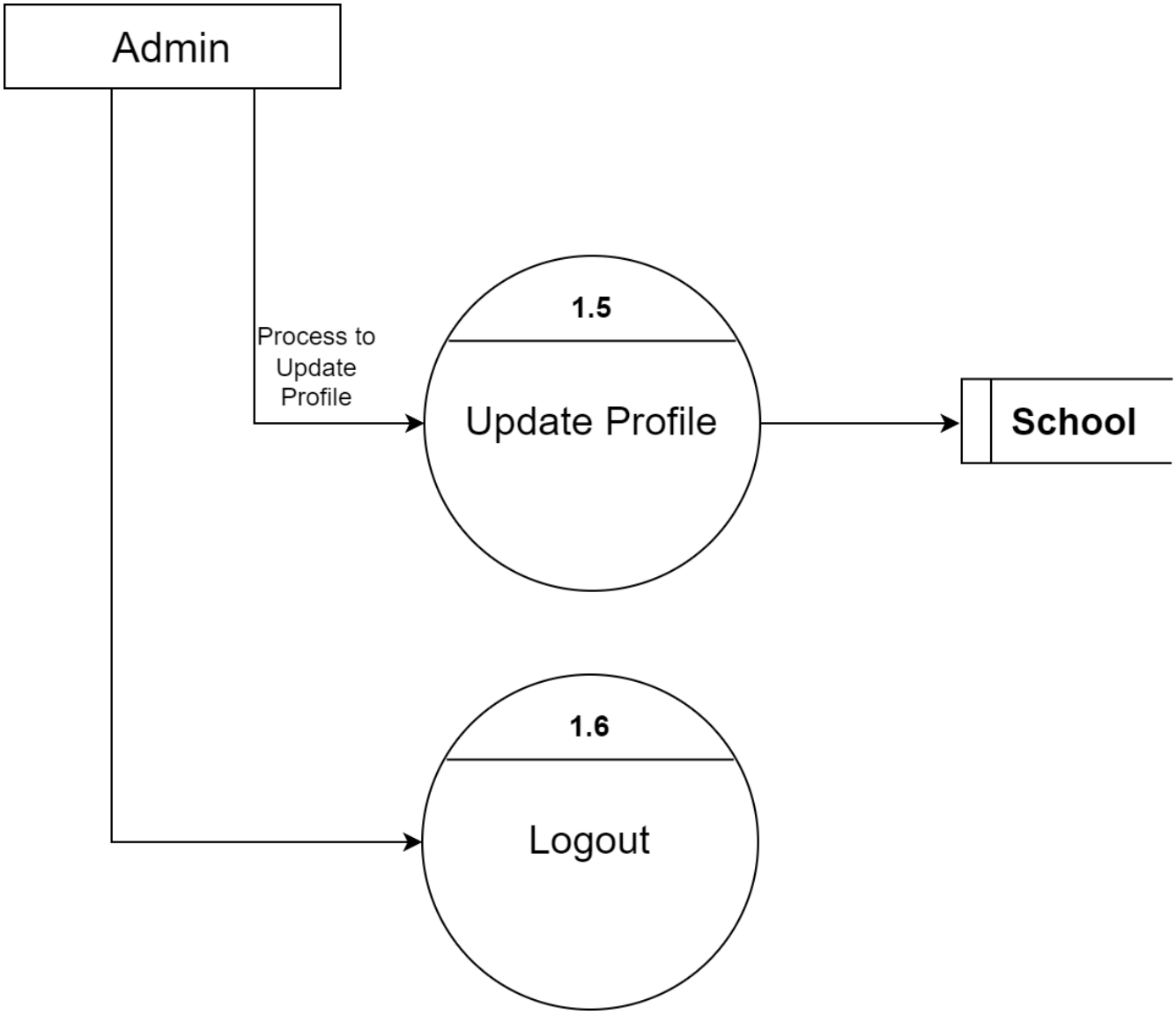
Context Level Diagram

0th Level :-

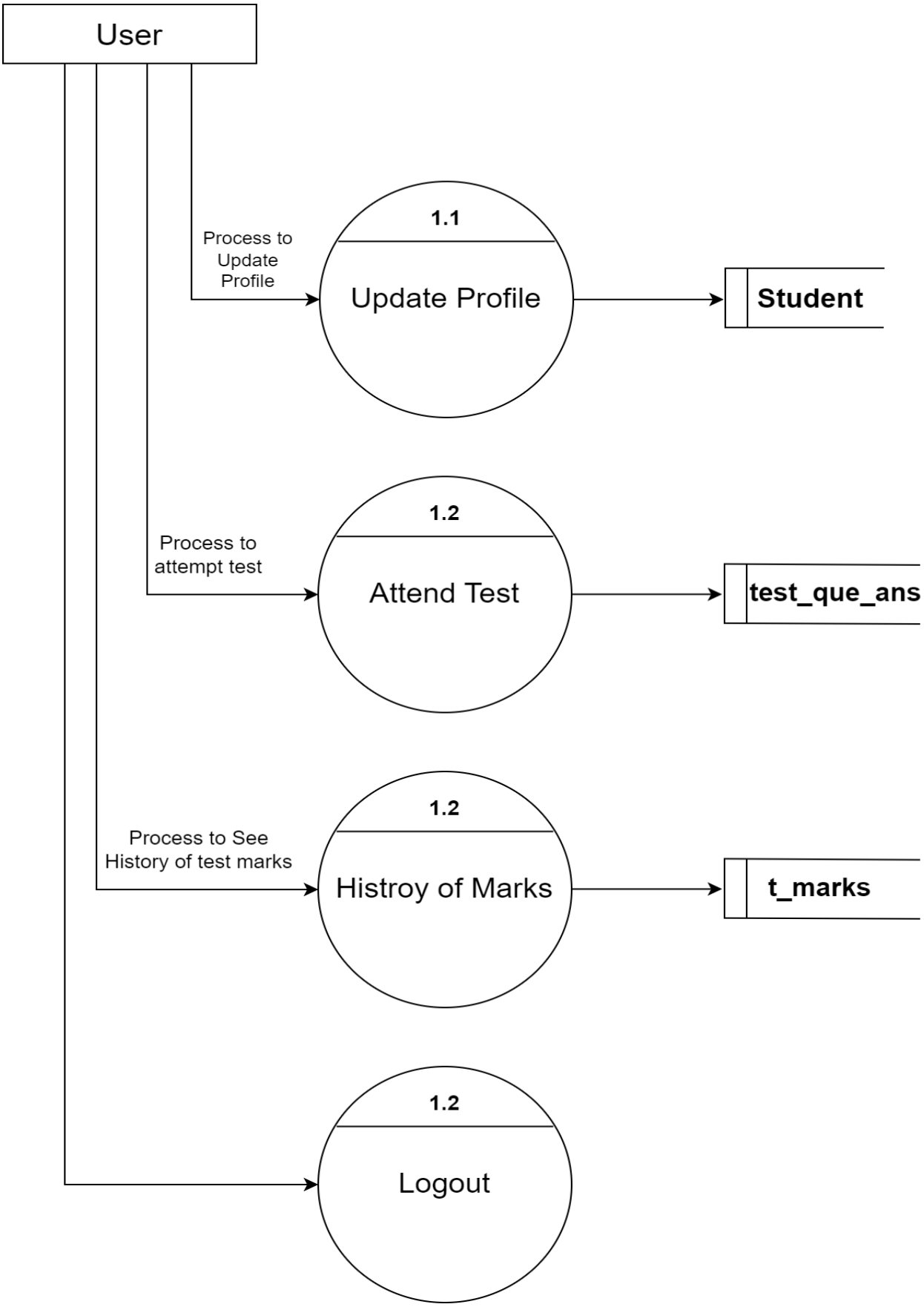


1th Level for Admin Side :-

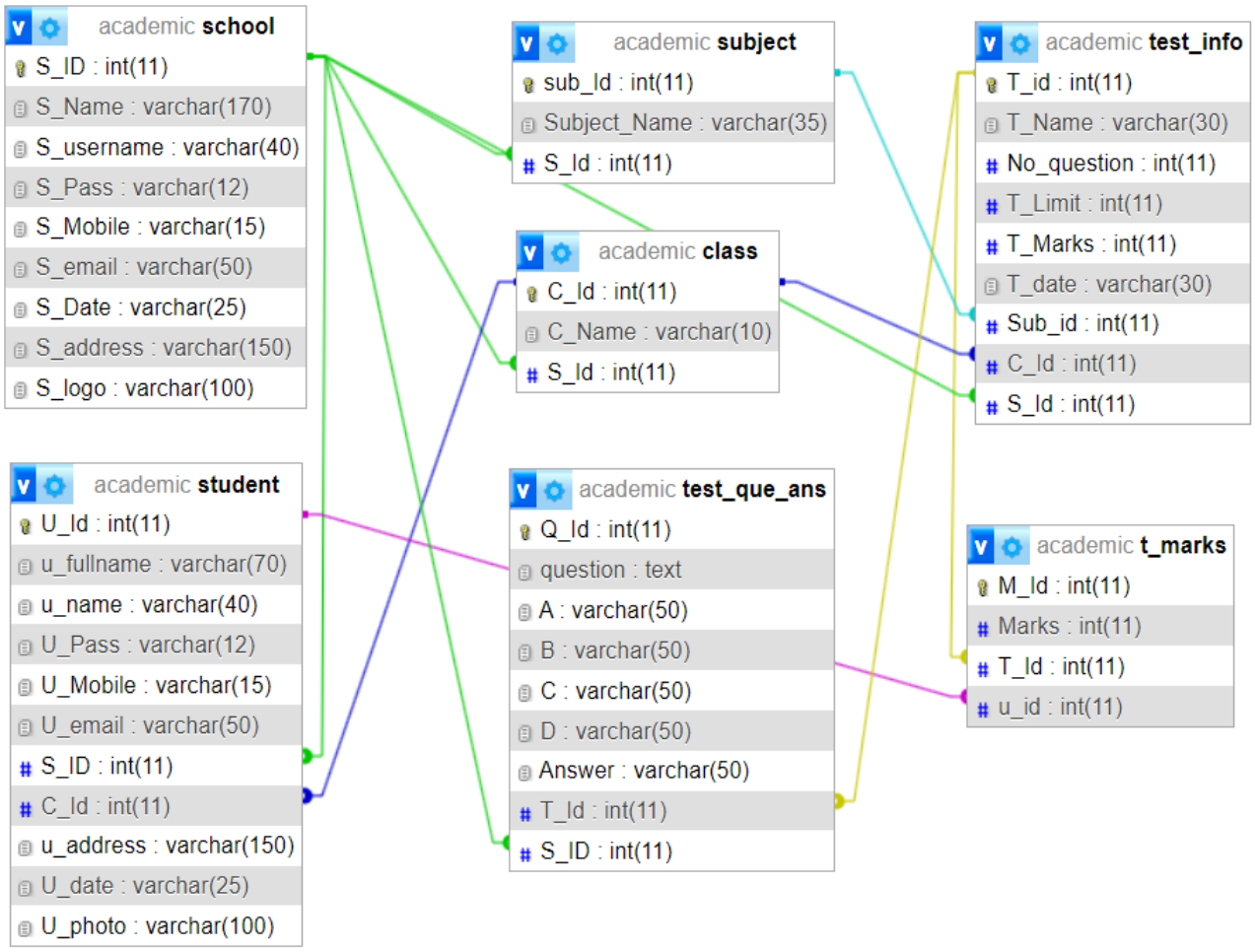



Admin
Process to Update Profile
1.5
Update Profile
School
1.6
Logout

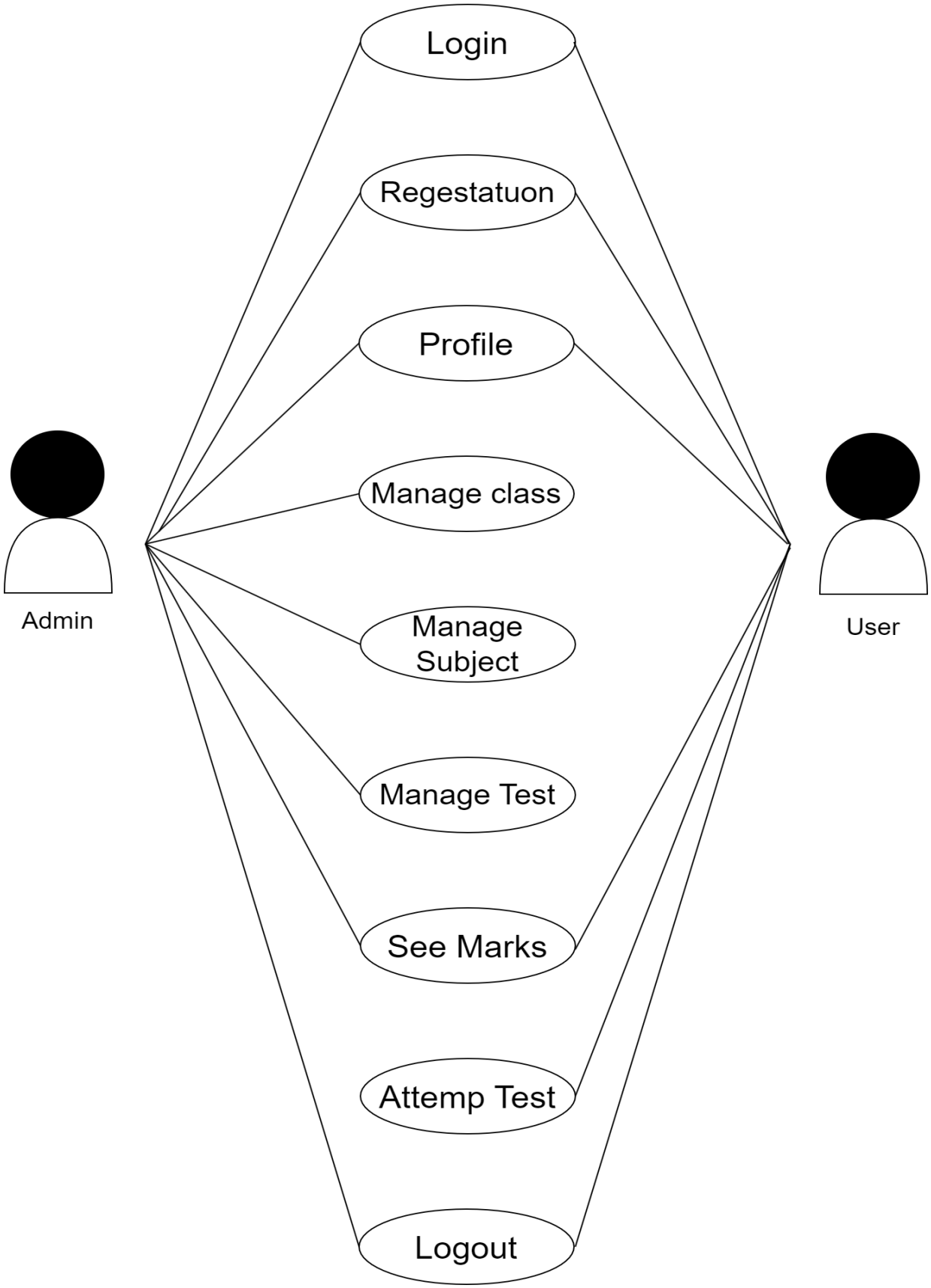
1th Level for User Side :-



ER Diagram

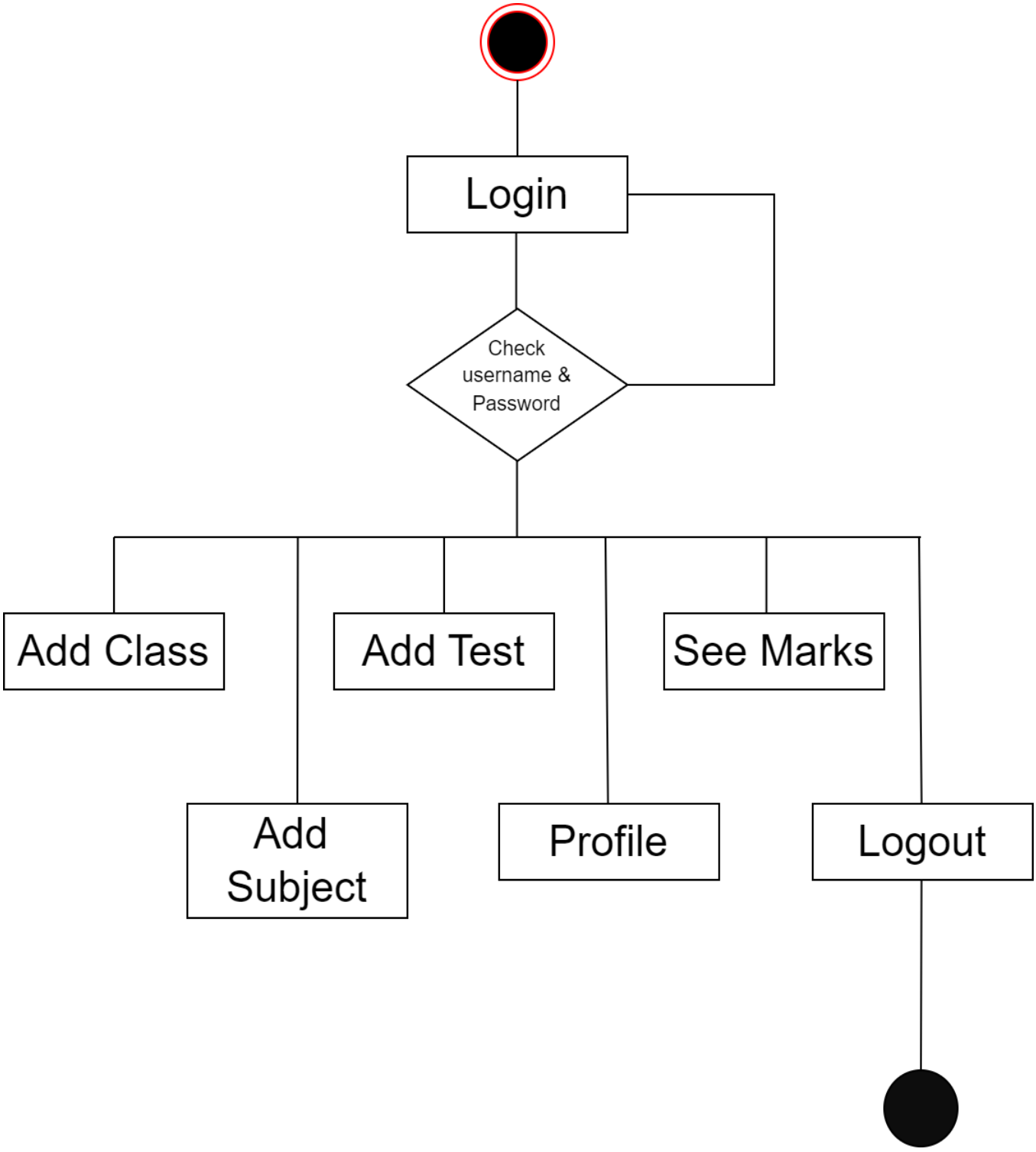

academic school
S_ID : int(11)
S_Name : varchar(170)
S_username : varchar(40)
S_Pass : varchar(12)
S_Mobile : varchar(15)
S_email : varchar(50)
S_Date : varchar(25)
S_address : varchar(150)
S_logo : varchar(100)
academic subject
sub_Id : int(11)
Subject_Name : varchar(35)
S_Id : int(11)
academic test_info
T_id : int(11)
T_Name : varchar(30)
No_question : int(11)
T_Limit : int(11)
T_Marks : int(11)
T_date : varchar(30)
Sub_id : int(11)
C_Id : int(11)
S_Id : int(11)
academic class
C_Id : int(11)
C_Name : varchar(10)
S_Id : int(11)
academic student
U_Id : int(11)
u_fullname : varchar(70)
u_name : varchar(40)
U_Pass : varchar(12)
U_Mobile : varchar(15)
U_email : varchar(50)
S_ID : int(11)
C_Id : int(11)
u_address : varchar(150)
U_date : varchar(25)
U_photo : varchar(100)
academic test_que_ans
Q_Id : int(11)
question : text
A : varchar(50)
B : varchar(50)
C : varchar(50)
D : varchar(50)
Answer : varchar(50)
T_Id : int(11)
S_ID : int(11)
academic t_marks
M_Id : int(11)
Marks : int(11)
T_Id : int(11)
u_id : int(11)

Usecase Diagram

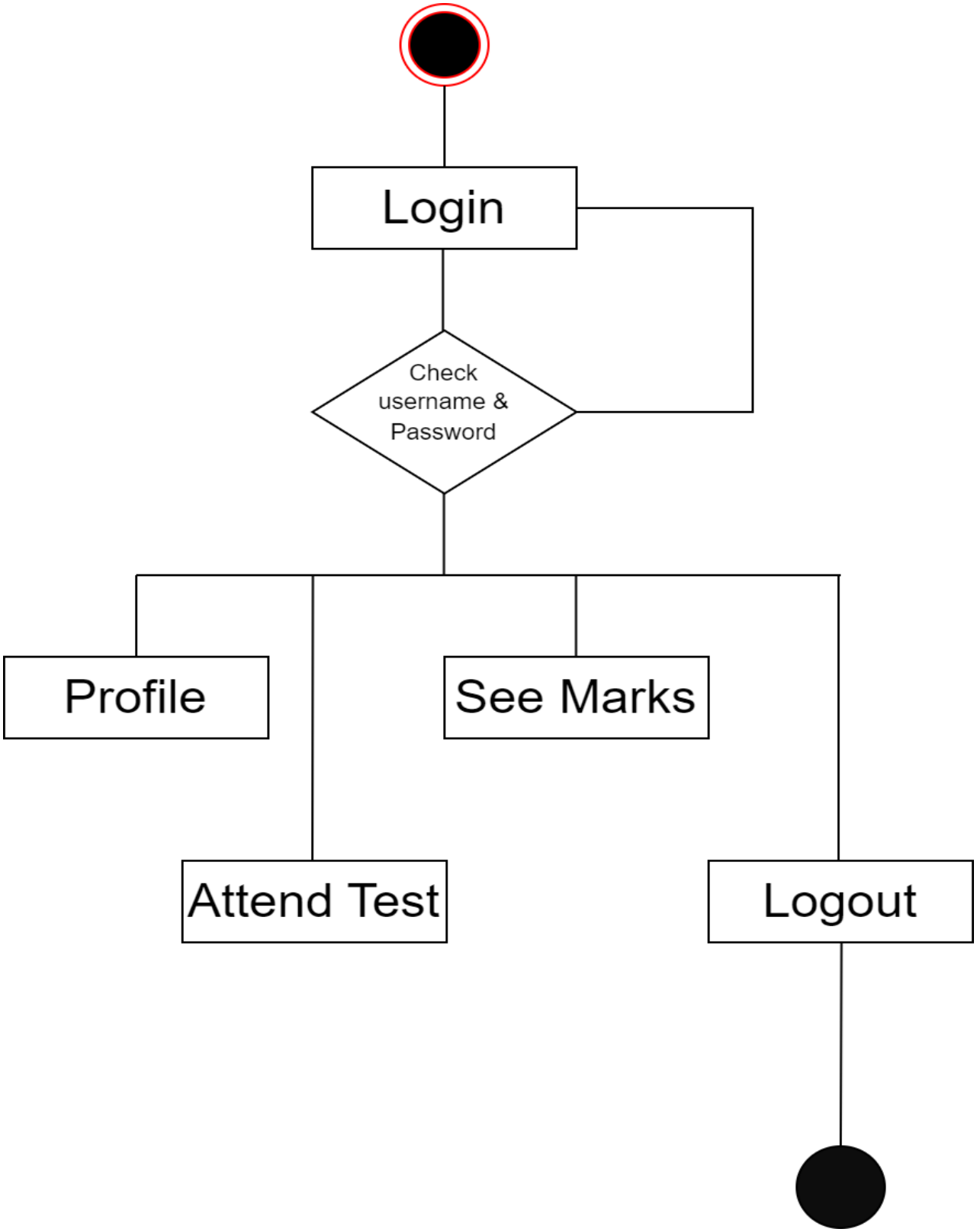


Activity Diagram

Admin Side

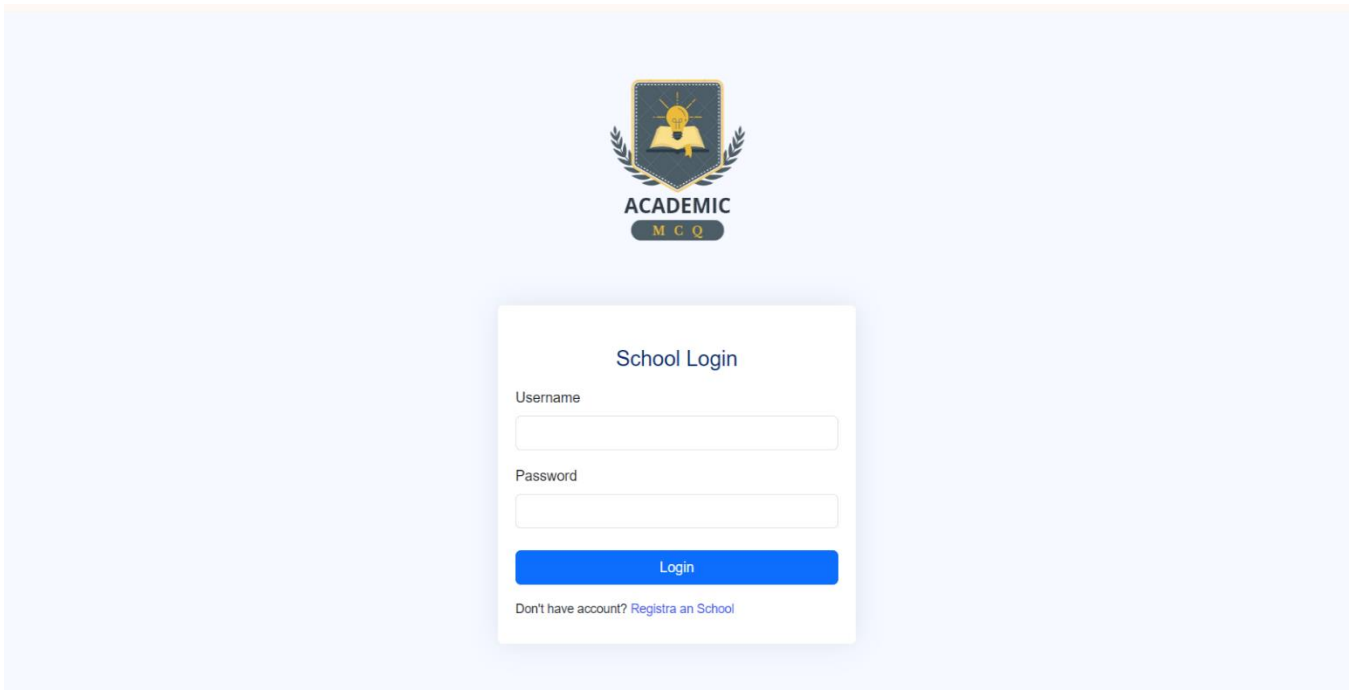


ScreenShots

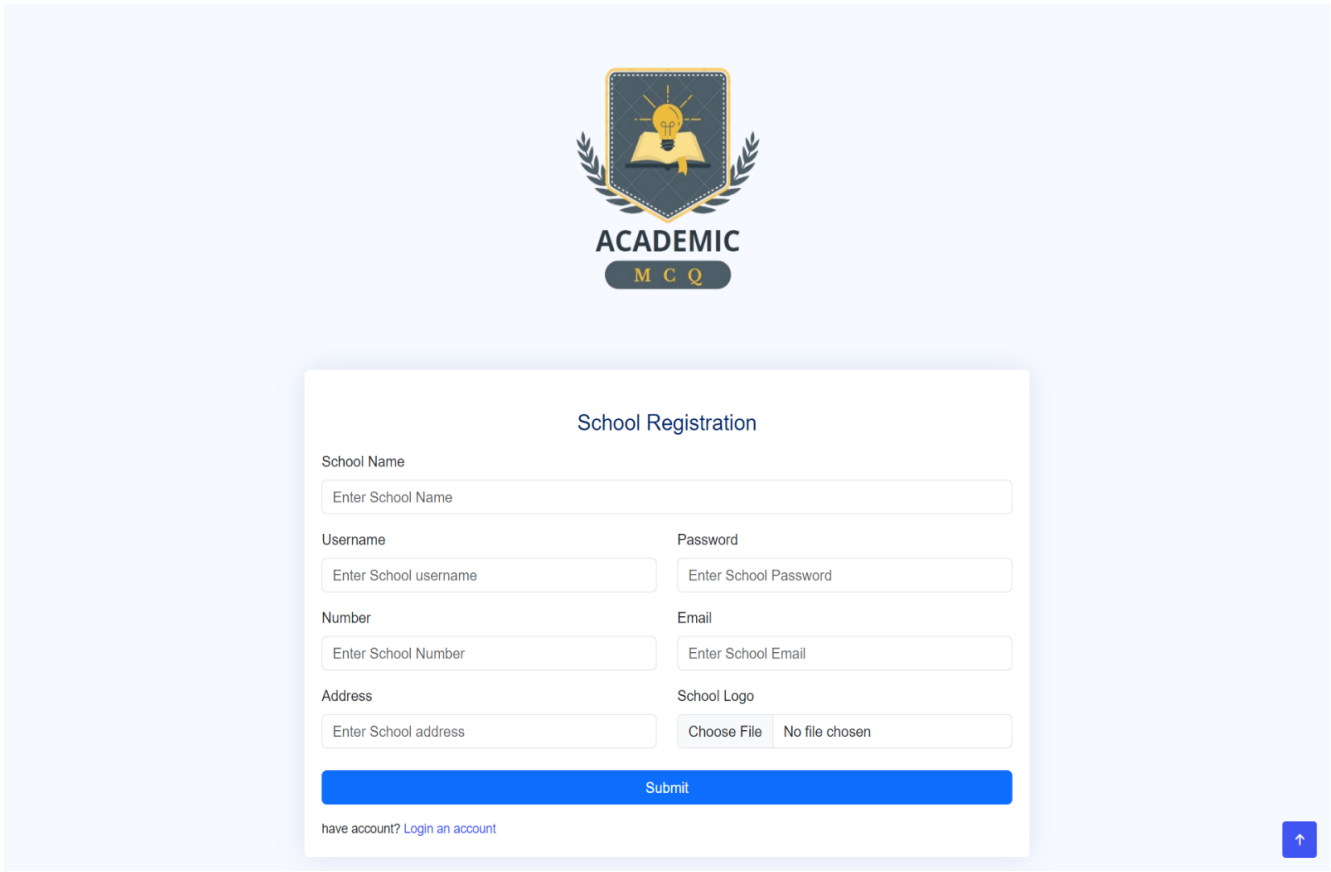


ScreenShots

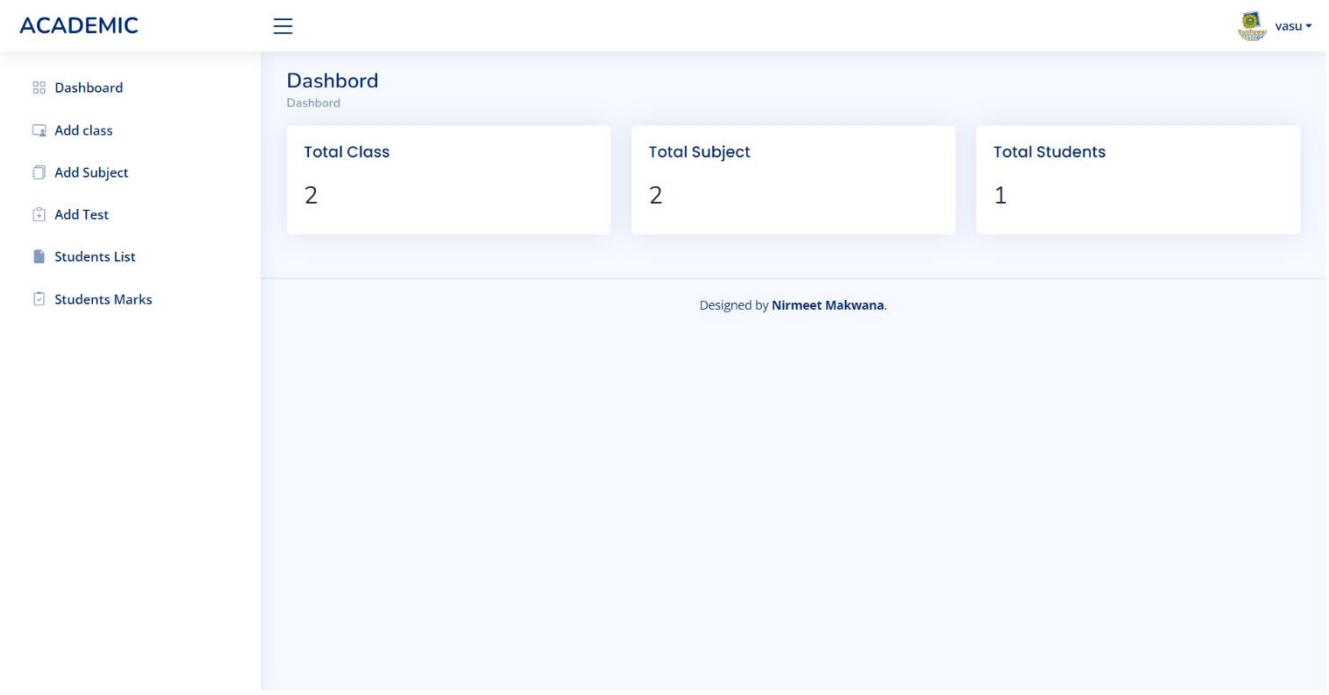
School Login Page



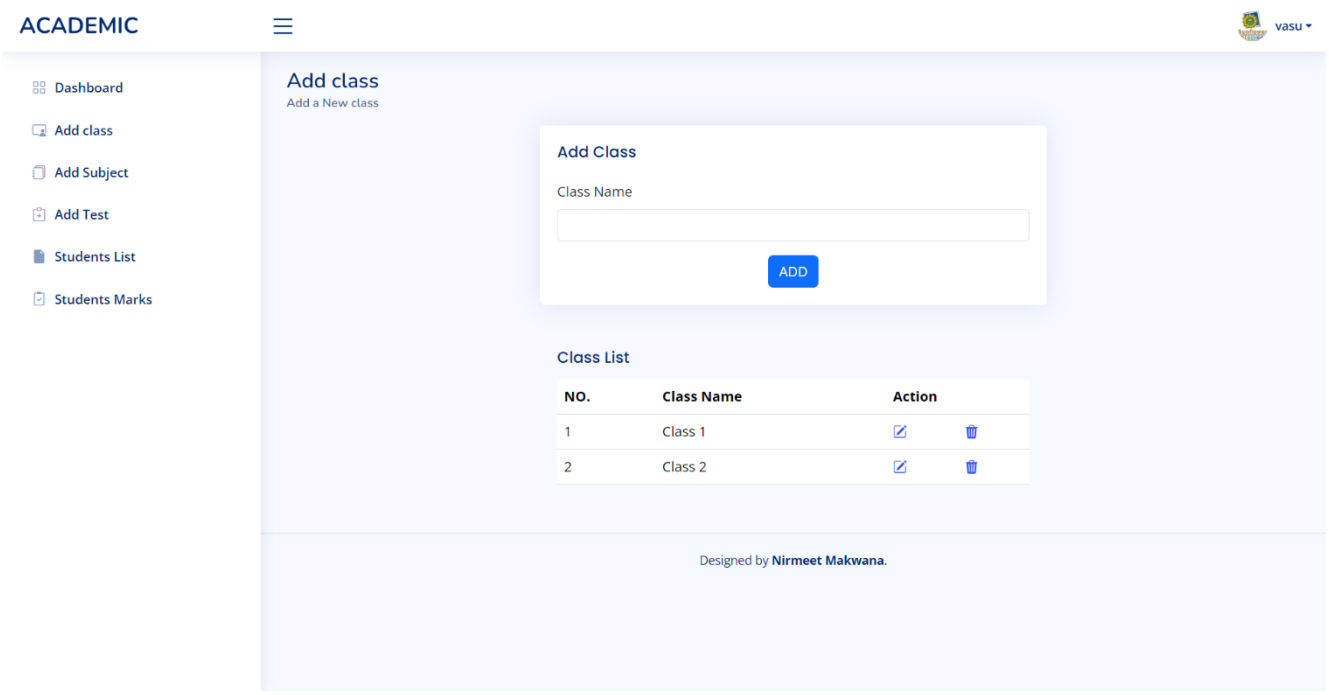
School Sign-up Page



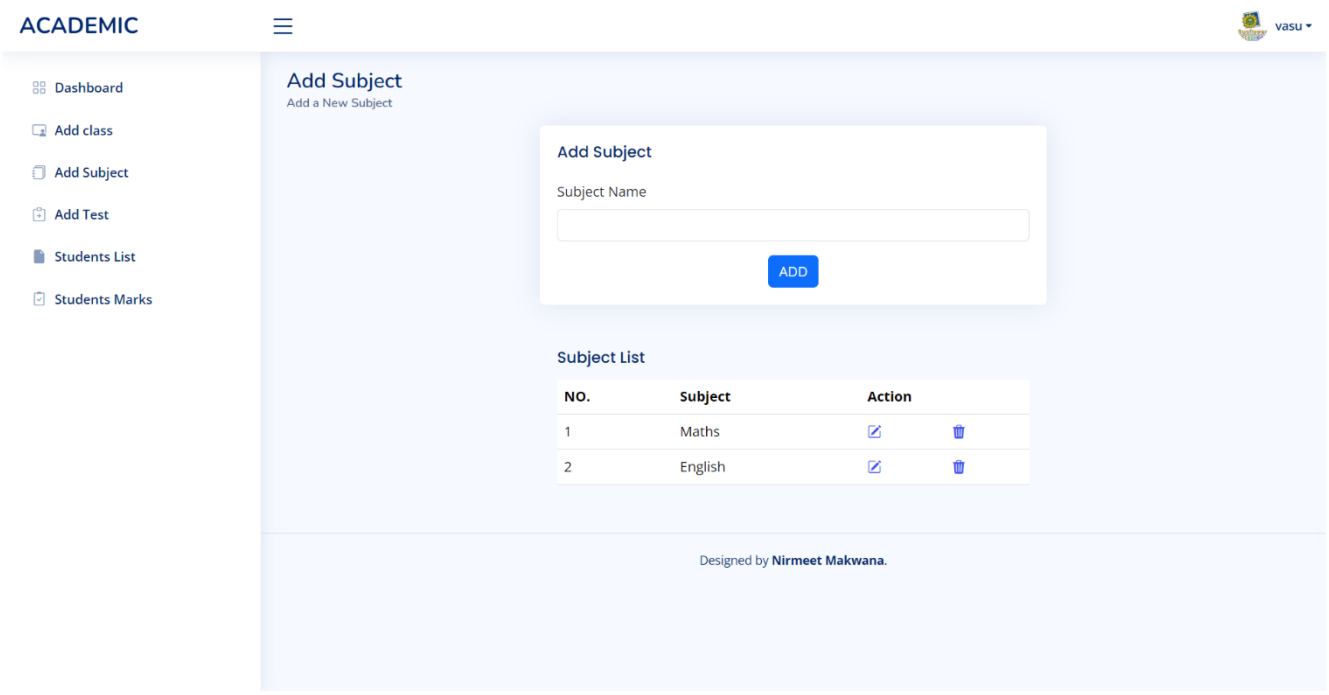
School Dashboard Page



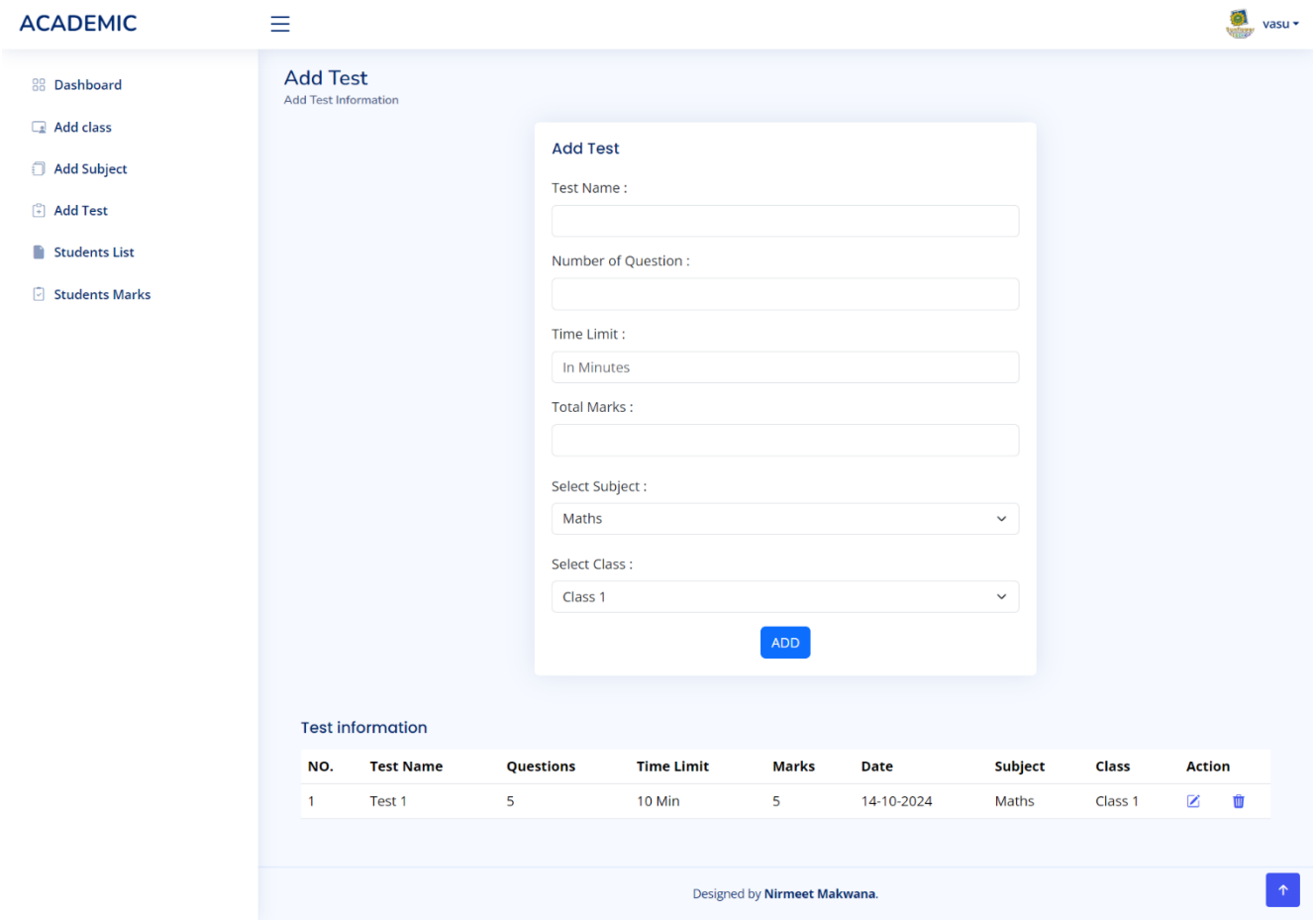
Add Class Page



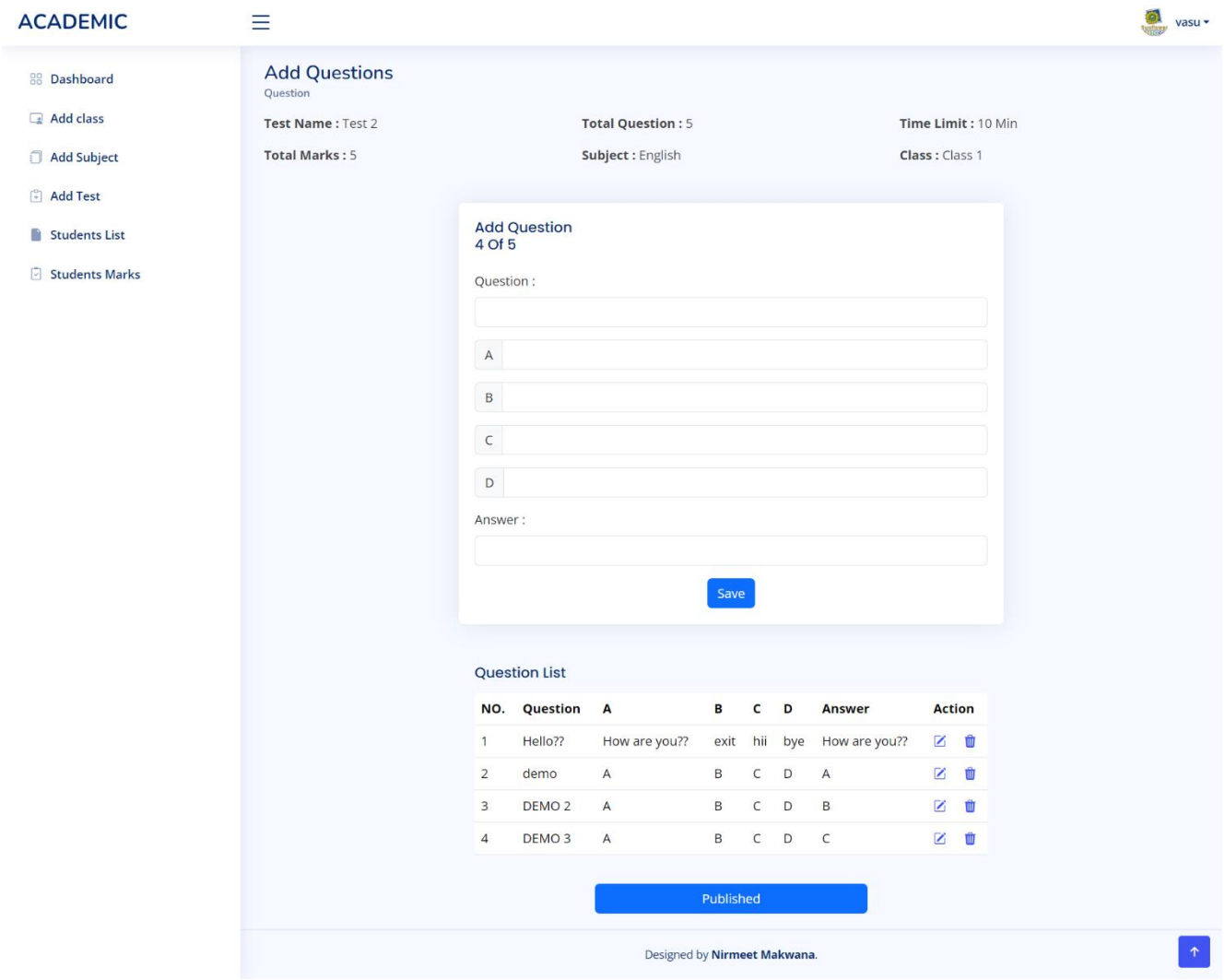
Add Subject Page



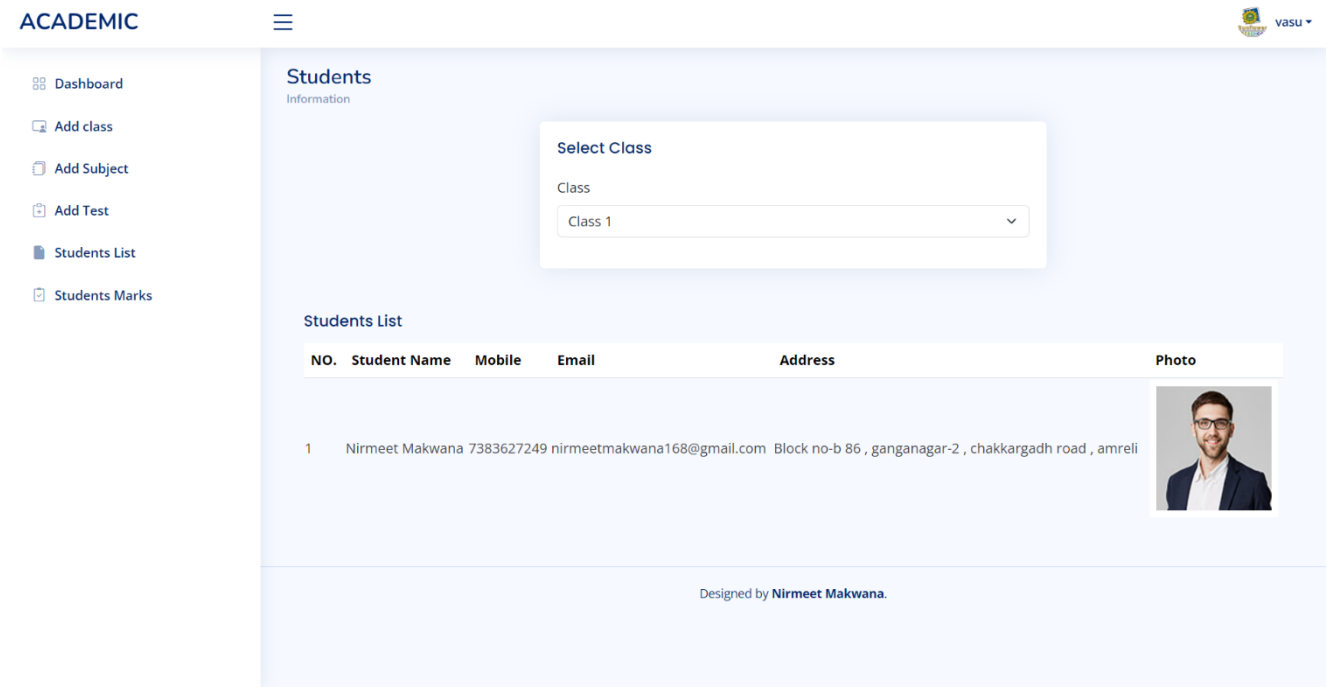
Add Test Page



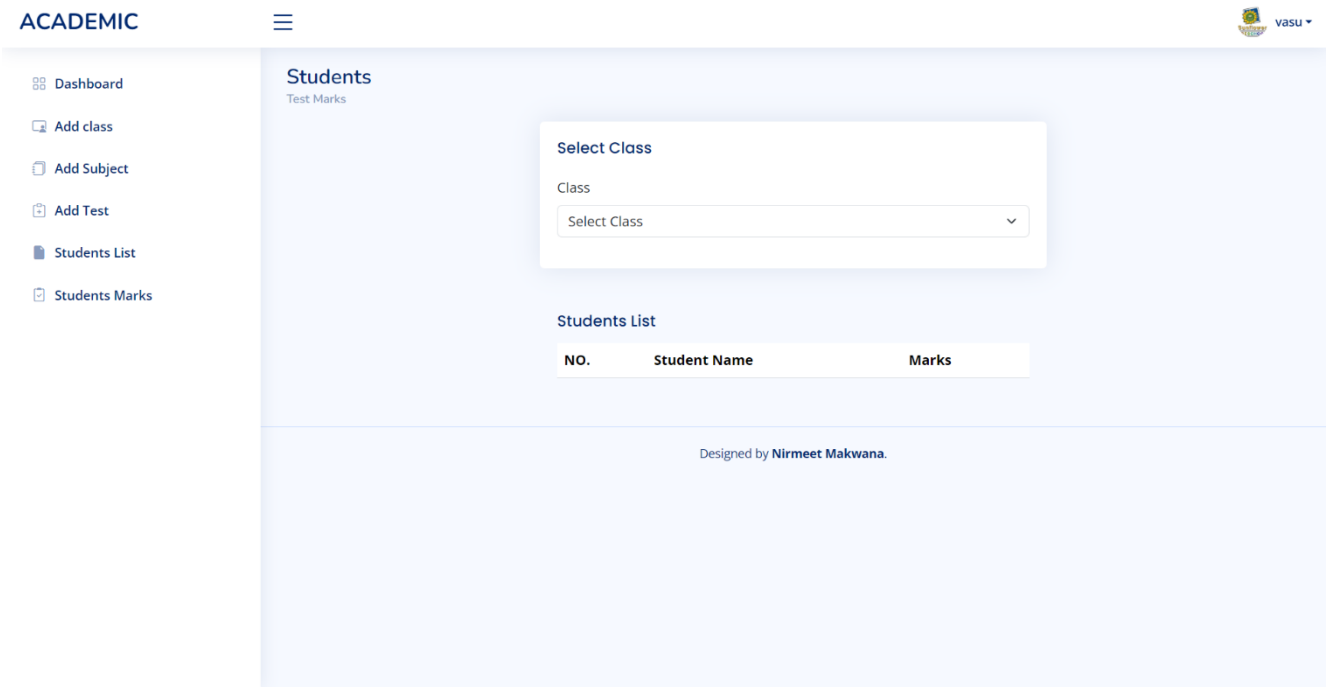
Add Question Answer Page



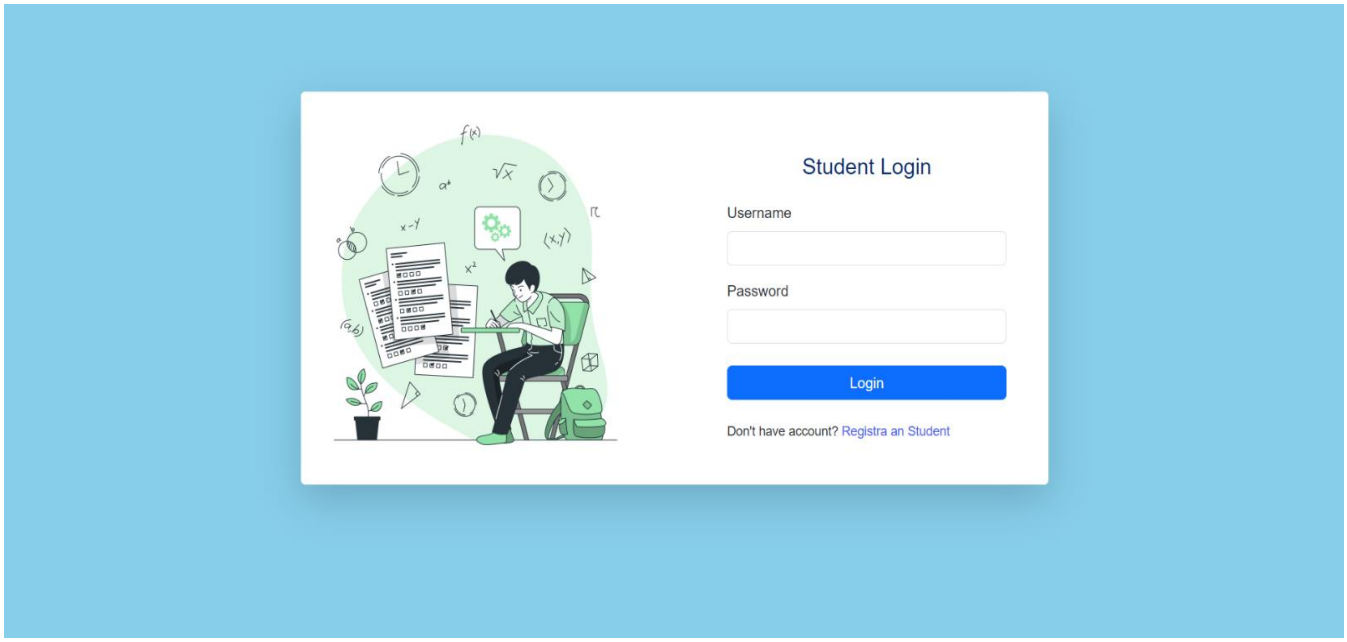
Student List Page



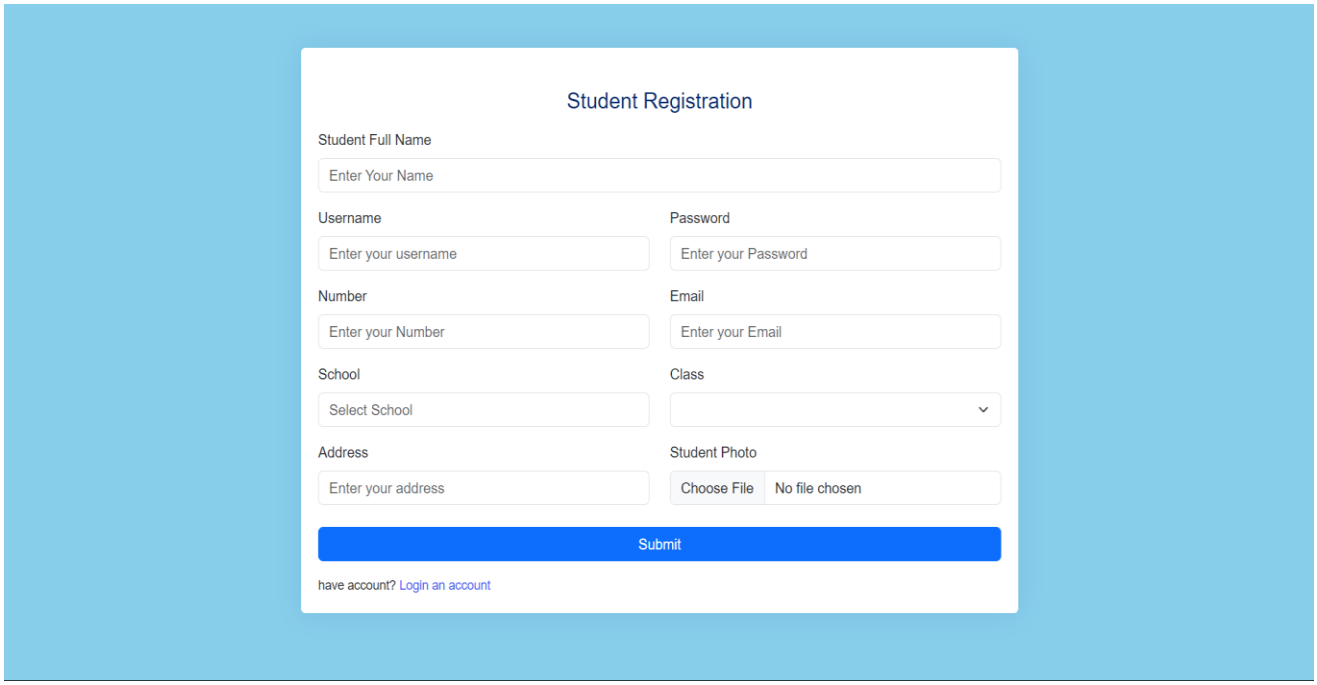
Student Marks Page



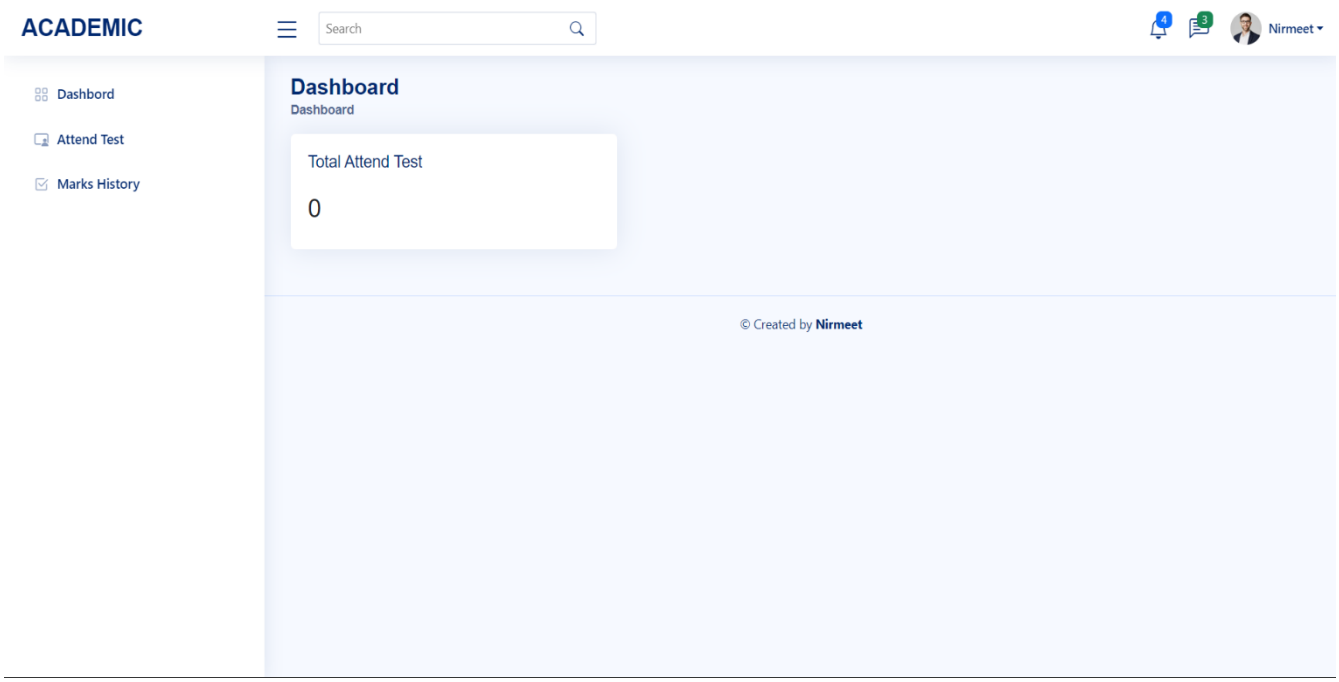
Student Page



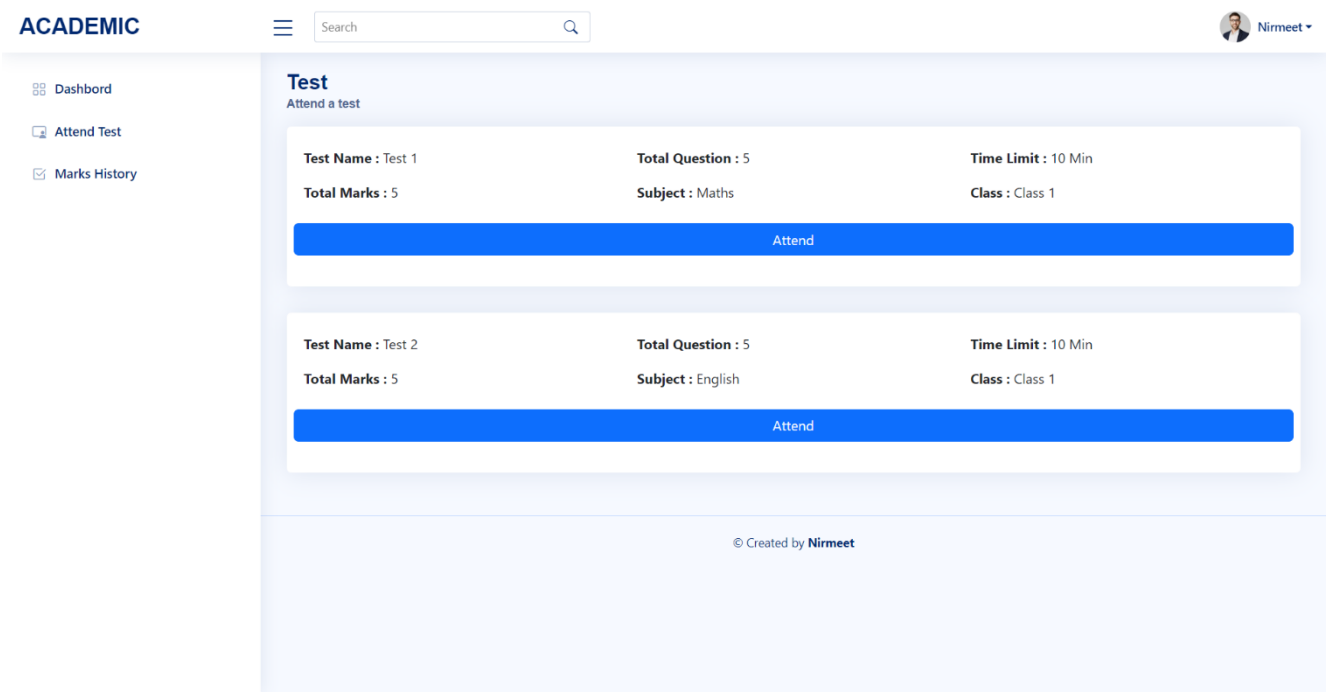
Student Sign-up Page



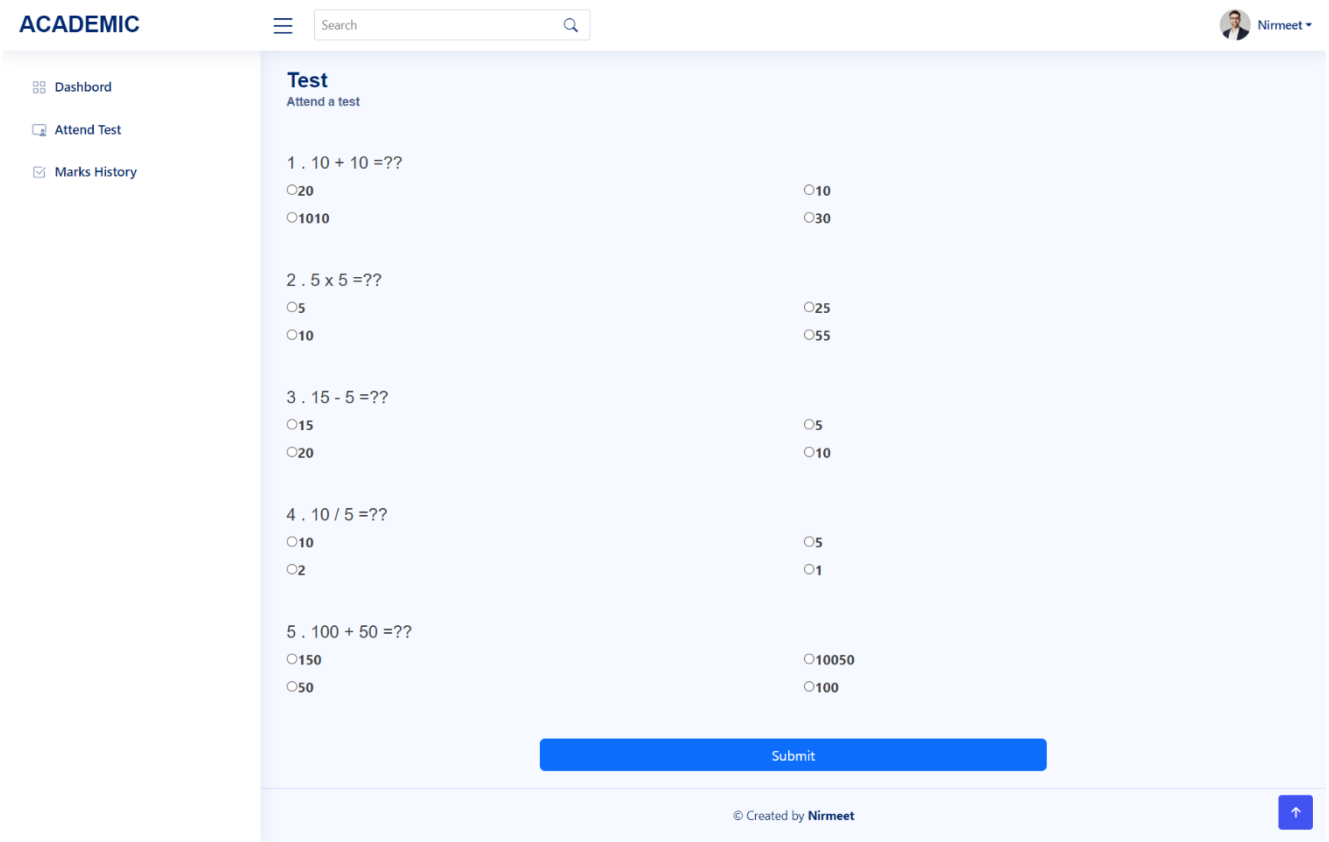
Student Dashboard Page



Select Test Page



Attend Test Page



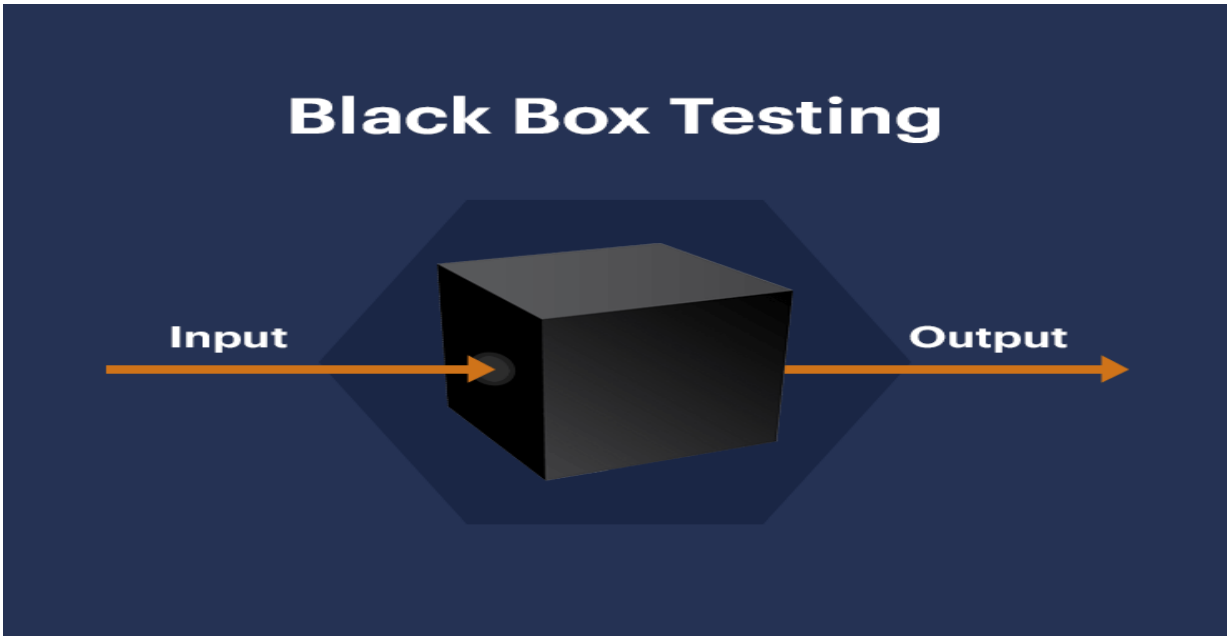
Testing

- **What is testing?**

  ⇨ Testing is a type of software testing that evaluates the overall functionality and performance of a complete and fully integrated software solution.

❖ **Black box testing:**

⇨ **Black box testing** is a software testing method where the tester evaluates the functionality of an application without knowing its internal code or structure.



- **White box testing:**

⇨ White box testing is a software testing method where the tester examines the internal logic and structure of the code.



- **Grey box testing:**

⇨ **Gray box testing** is a software testing method that combines black box and white box testing.



Test Case

Project Name → Online Examination System

Test Objective → Login Page

Test case Id	Test Step	Input Data	Expected Result	Actual Result	Exception Result
To Login 001	To check Username and password click login button	Enter Username and password	User should be go on page	User Login Successfully	Successful
To Login 002	To check Username and password click login button	Enter Invalid Username and password	User Should not be go on home page and comeback to login page	User not login	Unsuccessful
To Login 003	To check Username and password click login button	Enter Invalid Username and valid password	User should not be go on home page and comeback to login page	User not login	Unsuccessful
To Login 004	To check Username and password click login button	Enter Invalid name/shop name and invalid password	User should not be go on home page and comeback to login page	User not login	Unsuccessful

Advantages & Disadvantages

❖ Advantages :

✓ Students can take exams from any location with an internet connection, making it accessible for distance learners.

✓ Eliminates the need for physical exam centers, printing paper, and other logistics, reducing costs for institutions.

❖ Disadvantages :

✓ Unreliable internet connectivity, power outages, or server downtime can disrupt exams, leading to stress or unfair conditions for some students.

Future Scope

The future scope of an **Online Examination System** (OES) can be quite expansive, given the increasing reliance on digital tools in education and assessment.

Expanding digital examinations to eliminate the need for paper, making the process more eco-friendly.

Bibliography

1. www.google.com
2. PHP Manual
3. getbootstrap.com

CD/DVD

- Below items are included in CD/DVD:
  1. Project
  2. Database
  3. Documentation
  4. Screenshots



THANK YOU